Lecture 5: Solving Linear Equations via Cross-Filling

MATH203 Applied Linear Algebra

Motivation

Today's Goal

Learn to solve: $A\mathbf{x} = \mathbf{b}$ using the cross-filling method.

Why?

- Find all solutions (existence)
- Determine uniqueness
- Find basis for null space (relationships between columns)

Method: Decompose augmented matrix $(A \mid \mathbf{b})$ into rank-one pieces, extract simplified equations, solve backwards.

Complete Example: Solving $A\mathbf{x} = \mathbf{b}$

Example Setup

Solve:

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 4 \\ 9 \\ 13 \end{pmatrix}$$

Strategy: Form augmented matrix $(A \mid \mathbf{b})$, use cross-filling to simplify, then solve backwards.

Original Equation System

The matrix equation $A\mathbf{x} = \mathbf{b}$ represents three equations:

$$\begin{aligned} \text{eq1}: &\quad x_1 + 2x_2 + x_3 = 4 \\ \text{eq2}: &\quad 2x_1 + 4x_2 + x_3 = 9 \\ \text{eq3}: &\quad 3x_1 + 6x_2 + 2x_3 = 13 \end{aligned}$$

Goal: Use cross-filling to transform this into a simpler equivalent system.

Step 1: Form Augmented Matrix

$$(A \mid \mathbf{b})$$

$$\left(\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 2 & 4 & 1 & 9 \\ 3 & 6 & 2 & 13 \end{array}\right)$$

Each row represents one equation from our system.

Next: Choose a pivot to begin cross-filling.

Step 2: Choose First Pivot at (1,1)

$$\begin{array}{c} \color{red}{\text{pivot}} \\ \left(\begin{array}{ccccc} \boxed{1} & 2 & 1 & & 4 \\ 2 & 4 & 1 & & 9 \\ 3 & 6 & 2 & & 13 \end{array}\right) \end{array}$$

Pivot: entry $(1, 1) = 1$ (red)

Cross-filling formula:

$$(R_1 \mid \mathbf{b}_1) = \underbrace{\frac{\text{column } 1}{1}}_{\text{normalized column}} \cdot (\text{row } 1)$$

Step 3: First Decomposition Result

$$\underbrace{\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 2 & 4 & 1 & | & 9 \\ 3 & 6 & 2 & | & 13 \end{pmatrix}}_{(A|\mathbf{b})} = \underbrace{\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 2 & 4 & 2 & | & 8 \\ 3 & 6 & 3 & | & 12 \end{pmatrix}}_{(R_1|\mathbf{b}_1)\ \text{rank-one}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & | & 1 \\ 0 & 0 & -1 & | & 1 \end{pmatrix}}_{\text{Remainder}}$$

Key observation: Column 1 eliminated from Remainder!

Step 5: Equations from $(R_1 \mid \mathbf{b}_1)$

The matrix:

$$(R_1 \mid \mathbf{b}_1) = \begin{pmatrix} 1 & 2 & 1 & \mid & 4 \\ 2 & 4 & 2 & \mid & 8 \\ 3 & 6 & 3 & \mid & 12 \end{pmatrix}$$

Each row represents an equation:

$$\begin{aligned} \text{Row 1}: &\quad 1x_1 + 2x_2 + 1x_3 = 4 \quad (\mathbf{eq1}) \\ \text{Row 2}: &\quad 2x_1 + 4x_2 + 2x_3 = 8 \quad (\mathbf{2{\cdot}eq1}) \\ \text{Row 3}: &\quad 3x_1 + 6x_2 + 3x_3 = 12 \quad (\mathbf{3{\cdot}eq1}) \end{aligned}$$

Key: All rows = same equation (eq1). Only **one independent equation**!

Step 4: Understanding $(R_1 \mid \mathbf{b}_1)$

How did we get $(R_1 \mid \mathbf{b}_1)$?

$$(R_1 \mid \mathbf{b}_1) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \end{array}\right)$$

This is a **rank-one matrix**: all rows are multiples of row 1.

Step 6: Equations from Remainder

The matrix:

$$\text{Remainder} = \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right)$$

Each row represents an equation:

$$\begin{aligned} \text{Row 1}: &\quad 0 = 0 \quad (\text{actively zero: pivot row}) \\ \text{Row 2}: &\quad -x_3 = 1 \quad (\text{passively zero in col 1: eq2 - 2}\cdot\text{eq1}) \\ \text{Row 3}: &\quad -x_3 = 1 \quad (\text{passively zero in col 1: eq3 - 3}\cdot\text{eq1}) \end{aligned}$$

Key distinction:

- Row 1: actively became zero (we chose it as pivot row)
- Rows 2, 3: passively became zero in column 1 (cross-filling eliminated it)

Step 7: Understanding the Decomposition

We decomposed $(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) +$ Remainder:

Matrix	**Equations contained**	**Relation to original**
$(R_1 \mid \mathbf{b}_1)$	eq1 ($\times$3 rows)	eq1
Remainder	$-x_3 = 1$ ($\times$2 rows)	eq2 $-$ 2·eq1, eq3 $-$ 3·eq1

Simplified equation system:

$$\begin{aligned} \text{From } R_1: &\quad x_1 + 2x_2 + x_3 = 4 \\ \text{From Remainder}: &\quad x_3 = -1 \end{aligned}$$

This is equivalent to the original three equations!

Step 8: Continue Cross-Filling on Remainder

The Remainder still has structure. Choose second pivot at $(2,3) = -1$:

$$\begin{array}{c} \begin{array}{cccc} & & \text{pivot} & \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \end{array}$$

$$(R_2 \mid \mathbf{b}_2) = \frac{\text{column 3 of Remainder}}{-1} \cdot (\text{row 2 of Remainder})$$

Step 9: Second Decomposition Result

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & | & 1 \\ 0 & 0 & -1 & | & 1 \end{pmatrix}}_{\text{First Remainder}} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & -1 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}}_{(R_2|\mathbf{b}_2)\text{ rank-one}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}}_{\text{All zeros!}}$$

Cross-filling complete: $(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + (R_2 \mid \mathbf{b}_2)$

Step 10: Equations from $(R_2 \mid \mathbf{b}_2)$

The matrix:

$$(R_2 \mid \mathbf{b}_2) = \begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & -1 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

Each row represents an equation:

$$\begin{aligned} &\text{Row 1}: \quad 0 = 0 \quad \text{(inherited zero from Remainder)} \\ &\text{Row 2}: \quad x_3 = -1 \quad \text{(actively extracted: pivot row)} \\ &\text{Row 3}: \quad x_3 = -1 \quad \text{(passively copied: non-pivot row)} \end{aligned}$$

Note: Row 2 was chosen as pivot row; row 3 passively gets the same equation.

Step 10 (continued): Final Remainder Analysis

After extracting $(R_2 \mid \mathbf{b}_2)$, the final Remainder is all zeros:

$$\text{Final Remainder} = \begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

How did each row become zero?

- Row 1: was already 0 (inherited)
- Row 2: actively became 0 (we chose it as pivot row)
- Row 3: passively became 0 (row 3 - row 2, column 3 eliminated)

Cross-filling complete!

Step 11: The Simplified Equation System

From the cross-filling decomposition, we extracted:

From piece	Equation
$(R_1 \mid \mathbf{b}_1)$	$x_1 + 2x_2 + x_3 = 4$
	(pivot: x_1, free: x_2, x_3 not yet determined)
$(R_2 \mid \mathbf{b}_2)$	$x_3 = -1$
	(pivot: x_3)

Variable classification (from cross-filling):

- **Pivot variables**: x_1, x_3 (columns 1 and 3 where we chose pivots)
- **Free variable**: x_2 (column 2, non-pivot column)

Step 12: Backward Solving

Solve from the **simplest equation** backwards:

From equation 2:

$$x_3 = -1$$

Substitute into equation 1:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 4 \\ x_1 + 2x_2 + (-1) &= 4 \\ x_1 + 2x_2 &= 5 \\ x_1 &= 5 - 2x_2 \end{aligned}$$

Variable x_2 is not determined by any equation!

Step 13: Identify Pivot and Free Variables

From cross-filling, we chose pivots at:

- Column 1 (first pivot)
- Column 3 (second pivot)

Classification:

- **Pivot variables**: x_1, x_3 (determined by equations)
- **Free variable**: x_2 (can be chosen freely)

Pivot columns $=$ columns where we chose pivots.

Non-pivot columns $=$ remaining columns $\rightarrow$ free variables.

Step 14: General Solution

Let $x_2 = t$ (free parameter).

From our backward solving:

$$x_1 = 5 - 2t$$
$$x_2 = t$$
$$x_3 = -1$$

General solution:

$$\mathbf{x} = \begin{pmatrix} 5 - 2t \\ t \\ -1 \end{pmatrix}$$

This describes **infinitely many solutions** (one for each value of t).

Step 15: Decompose into Particular + Null Space

Rewrite the general solution:

$$\mathbf{x} = \begin{pmatrix} 5-2t \\ t \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

Two parts:

- **Particular solution** $\mathbf{x}_p = \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix}$ (when $t = 0$)
- **Null space direction** $\mathbf{v} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ (coefficient of t)

Check: $A\mathbf{v} = \mathbf{0}$ (verify this is in null space).

Step 16: Verification (Particular Solution)

Verify particular solution: $A\mathbf{x}_p = \mathbf{b}$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 5+0-1 \\ 10+0-1 \\ 15+0-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \\ 13 \end{pmatrix}$$

✓

Step 16: Verification (Null Space Vector) (continued)

Verify null space vector: $A\mathbf{v} = \mathbf{0}$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2+2+0 \\ -4+4+0 \\ -6+6+0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\checkmark$

Both parts of the solution are verified!

Summary of the Example

What we did:

1. Formed augmented matrix $(A \mid \mathbf{b})$
2. Used cross-filling to decompose into rank-one pieces
3. Extracted simplified equations from each piece
4. Solved backwards from simplest equation
5. Identified free variables
6. Wrote general solution = particular + null space

Key insight: Cross-filling transforms the original equation system into an **equivalent simpler system** where variables decrease step-by-step.

Why Variables Decrease

Observation: Variables Decrease with Each Piece

In our example:

Piece	**Active variables**	**Count**
Original $(A \mid \mathbf{b})$	x_1, x_2, x_3	3
After extracting R_1 Remainder	x_3 only	1
After extracting R_2 Final Remainder	none	0

Pattern: Each cross-filling step eliminates at least one variable from the remainder.

Why Does This Happen?

Cross-filling structure:

$$(R_i \mid \mathbf{b}_i) = \frac{\text{column } j}{a_{kj}} \cdot (\text{row } k)$$

- All rows of $(R_i \mid \mathbf{b}_i)$ are multiples of row k
- When we subtract $(R_i \mid \mathbf{b}_i)$ from the matrix, column j becomes zero in the remainder
- Column j is **eliminated** from future consideration

Result: Each step reduces the number of active columns (variables) by at least one.

Consequence: Backward Solving Works

Since variables decrease:

- Last piece $(R_r \mid \mathbf{b}_r)$ has **fewest variables**
- Second-to-last has one more variable
- $\vdots$
- First piece has most variables

Backward solving strategy:

1. Start from the last equation (fewest variables)
2. Solve for pivot variable(s)
3. Substitute into previous equation
4. Repeat until all pivot variables determined

Existence and Uniqueness

When Does $A\mathbf{x} = \mathbf{b}$ Have a Solution?

Key question: After cross-filling $(A \mid \mathbf{b})$, what determines existence of solutions?

Strategy: Always choose pivots **inside** A (not in $\mathbf{b}$ column).

Two possible outcomes:

1. **$\mathbf{b}$ column clears to zero** $\Rightarrow$ solution exists
2. **$\mathbf{b}$ column has residue after A clears** $\Rightarrow$ no solution

Let's see a concrete **no-solution example**...

Example: A System with NO Solution

Same A matrix, but different $\mathbf{b}$:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 4 \\ 9 \\ 14 \end{pmatrix}$$

Note: Changed last entry from 13 to 14.

Question: Can we solve $A\mathbf{x} = \mathbf{b}$?

Method: Cross-fill $(A \mid \mathbf{b})$ with pivots **only in A's columns**.

No-Solution Example: Initial Augmented Matrix

$$(A \mid \mathbf{b}) = \begin{pmatrix} 1 & 2 & 1 & \mid & 4 \\ 2 & 4 & 1 & \mid & 9 \\ 3 & 6 & 2 & \mid & 14 \end{pmatrix}$$

Cross-filling rule: Choose pivots **only in columns 1, 2, 3** (inside A).

Number of cross-fillings we can do on A $= \mathrm{rank}(A)$.

For this A: $\mathrm{rank}(A) = 2$ (we'll see this as we go).

No-Solution Step 1: First Cross-Filling

Choose pivot at (1,1):

$$\underbrace{\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 2 & 4 & 1 & | & 9 \\ 3 & 6 & 2 & | & 14 \end{pmatrix}}_{(A|\mathbf{b})} = \underbrace{\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 2 & 4 & 2 & | & 8 \\ 3 & 6 & 3 & | & 12 \end{pmatrix}}_{(R_1|\mathbf{b}_1)} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & | & 1 \\ 0 & 0 & -1 & | & 2 \end{pmatrix}}_{\text{Remainder}}$$

Observe:

- Column 1 cleared (zeros tattooed): rows 2 and 3, column 1
- Row 1 cleared (pivot row)
- **b** column changed: $14 \to 12 + 2$

No-Solution Step 2: Second Cross-Filling

Choose pivot at (2,3) in Remainder:

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & | & 1 \\ 0 & 0 & -1 & | & 2 \end{pmatrix}}_{\text{First Remainder}} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & \text{-1} \\ 0 & 0 & 1 & | & \text{-1} \end{pmatrix}}_{(R_2|\mathbf{b}_2)} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & \boxed{3} \end{pmatrix}}_{\text{Second Remainder}}$$

Critical observation:

- A part completely cleared (zeros tattooed everywhere in columns 1-3)
- Row 2 cleared (pivot row)
- **b** column has residue $\boxed{3} \neq 0$!

No-Solution: Why Residue Means No Solution

Second Remainder row 3:

$$\begin{pmatrix} 0 & 0 & 0 & | & \boxed{3} \end{pmatrix}$$

This represents the equation:

$$0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 3$$

$$\Downarrow$$

$$0 = 3 \quad \text{(CONTRADICTION!)}$$

Why did this happen?

- We exhausted all pivots in A ($\text{rank}(A) = 2$, did 2 cross-fillings)
- A part completely cleared
- But $\mathbf{b}$ column still has residue (3 left over)
- This means $\mathbf{b} \notin \text{Col}(A)$

No-Solution: Cannot Fix by More Cross-Filling

Can we do another cross-filling to clear the residue?

Second Remainder:

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{3} \end{array}\right)$$

Problem: The only non-zero entry is in the $\mathbf{b}$ column!

If we choose pivot at (3,4) (in $\mathbf{b}$ column):

$$(R_3 \mid \mathbf{b}_3) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \left(\begin{array}{ccc|c} 0 & 0 & 0 & 3 \end{array}\right) = \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{array}\right)$$

Still gives equation $0 = 3$! Cannot avoid contradiction.

No-Solution: The Key Insight

Rank interpretation:

- $\text{rank}(A) = 2$: We can do **2 cross-fillings** on A
- After 2 cross-fillings, A part is completely zero
- If $\mathbf{b}$ column has residue, $\text{rank}(A \mid \mathbf{b}) = 3 > 2$
- This extra "rank" forces a contradiction row

General principle:

$\text{rank}(A \mid \mathbf{b}) > \text{rank}(A)$

$\Updownarrow$

$\mathbf{b}$ has residue after A clears

$\Updownarrow$

Contradiction row $0 = c$ ($c \neq 0$) appears

$\Updownarrow$

No solution

Existence Criterion

Theorem (Existence)
$A\mathbf{x} = \mathbf{b}$ has a solution if and only if:

$$\mathrm{rank}(A \mid \mathbf{b}) = \mathrm{rank}(A)$$

Interpretation:

- If adding $\mathbf{b}$ as a column doesn't increase rank $\Rightarrow$ $\mathbf{b}$ is already in the span of A's columns
- Equivalently: $\mathbf{b} \in \mathrm{Col}(A)$

Uniqueness Criterion

If a solution exists, is it unique?

Theorem (Uniqueness)
If $A\mathbf{x} = \mathbf{b}$ has a solution, it is **unique** if and only if:

$$\text{Null}(A) = \{\mathbf{0}\}$$

Equivalently:

- No free variables
- $\text{rank}(A) = n$ (number of columns)

Proof idea: If $\mathbf{x}_1, \mathbf{x}_2$ both solve $A\mathbf{x} = \mathbf{b}$, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.

The Four Cases

	$\textbf{rank}(A) = n$ (no free vars)	$\textbf{rank}(A) < n$ (has free vars)
$\textbf{rank}(A \mid \mathbf{b}) = \textbf{rank}(A)$ (consistent)	**Unique** solution	**Infinitely many** solutions
$\textbf{rank}(A \mid \mathbf{b}) = \textbf{rank}(A) + 1$ (inconsistent)	**No solution**	**No solution**

Our example: $\text{rank}(A) = 2$, $\text{rank}(A \mid \mathbf{b}) = 2$, $n = 3 \Rightarrow$ infinitely many solutions.

Finding Null Space Basis

Definition: Null Space

Definition (Null Space)
For a matrix $A \in \mathbb{R}^{m \times n}$:

$$\mathrm{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Meaning: Vectors that get mapped to zero.

Interpretation: Relationships between the columns of A.

If $\mathbf{v} \in \mathrm{Null}(A)$, then $v_1\mathbf{a}_1 + \cdots + v_n\mathbf{a}_n = \mathbf{0}$ (linear dependence).

Example: Finding Null Space Basis for Our Matrix

Solve $A\mathbf{x} = \mathbf{0}$:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}$$

Form $(A \mid \mathbf{0})$ and cross-fill (same steps as before, but with $\mathbf{b} = \mathbf{0}$):

Simplified system:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 0 \\ x_3 &= 0 \end{aligned}$$

Backward Solving for Null Space

From the simplified system:

$$\begin{aligned} x_3 &= 0 \\ x_1 + 2x_2 + 0 &= 0 \quad \Rightarrow \quad x_1 = -2x_2 \end{aligned}$$

Free variable: $x_2 = t$

General null space vector:

$$\mathbf{x} = \begin{pmatrix} -2t \\ t \\ 0 \end{pmatrix} = t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

Basis for Null(A): $\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right\}$

General Procedure: Finding Null Space Basis

Proposition
To find a basis for Null(A):

1. Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling
2. Identify free variables (non-pivot columns)
3. For each free variable: set it to 1, others to 0, solve for pivots
4. Each setting gives one basis vector

Dimension of Null(A) = number of free variables.

Rank-Nullity Theorem

Counting Pivot and Free Variables

For an $m \times n$ matrix A:

- Total columns: n
- Pivot columns (from cross-filling): $r = \text{rank}(A)$
- Non-pivot columns (free variables): $n - r$

But non-pivot columns = dimension of null space!

$$\dim(\text{Null}(A)) = n - r$$

Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)
For any $m \times n$ matrix A:

$$\boxed{\text{rank}(A) + \dim(\text{Null}(A)) = n}$$

Proof: Every column is either:

- A pivot column (contributes to rank), or
- A non-pivot column (contributes one basis vector to null space)

These counts partition the n columns.

Also written: $\text{rank}(A) + \text{nullity}(A) = n$

Example Verification

Our 3×3 example:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}$$

- $n = 3$ (three columns)
- $\text{rank}(A) = 2$ (two pivot columns: 1 and 3)
- $\dim(\text{Null}(A)) = 1$ (one free variable: x_2)

Check: $2 + 1 = 3$ ✓

Summary

What We Learned

Cross-filling method for solving $A\mathbf{x} = \mathbf{b}$:

1. Decompose $(A \mid \mathbf{b})$ into rank-one pieces
2. Each piece gives one simplified equation
3. Variables decrease with each piece
4. Solve backwards from simplest equation

Key results:

- Existence: $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$
- Uniqueness: $\text{Null}(A) = \{\mathbf{0}\}$
- General solution $=$ particular $+$ null space
- Rank-Nullity: $\text{rank}(A)+$ nullity $= n$

Next Lecture

Lecture 6: Four Fundamental Subspaces

We'll explore:

- Column space and null space (from today)
- Row space and left null space
- Orthogonality relationships: $\mathrm{Null}(A) \perp \mathrm{Row}(A)$
- The fundamental theorem of linear algebra